

RMO(ORISSA) – 2003

Max Mark – 100

Time – 3 Hrs

Instructions:

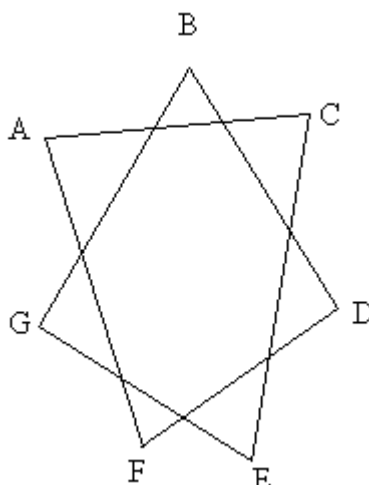
- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer as many questions as you can.
- Group A Q. 1 to Q.10 carry 4 marks each & Group B Q. 11 to Q. 20 carry 6 marks each.
- Bonus question each carry 10 marks will taken into account in case of tie.

Group – A

1. If one thing sold in Rs x it gives 15% loss, but if it sold in Rs y then it gives 15% gain. Then find the ratio of y to x.
2. Solve for x and y : $(x - y + 3)^2 + (x^2 + y - 3)^2 = 0$.
3. There are 35 coins with ten numbers of students. Out of ten students at least one has exactly 1 coin, at least one student having exactly 2 coins and at least one student having exactly 3 coins. Then prove that at least one student having 5 or more than 5 coins.
4. Out of four whole numbers taking three at a time their sum is as follows: 180, 197, 208 and 222. Then find the largest number amongst them.
5. Lengths of sides of a triangle are integers. If perimeter of triangle is 8 then find its area..
6. In a marriage party 13 pairs (husband & wife) joined. Each male handshakes with all other female except his wife. No female handshake with females. Find the number of handshakes between them.
7. Find the 2000th letter in the sequence ABCDEDCBAABCDEDCBAABCDEDCBAABC...
8. If x and y are real numbers such that $\frac{4^x}{2^{x+y}} = 8$ and $\frac{9^{x+y}}{3^{5y}} = 243$, then find value of xy.
9. One right angled triangle is drawn by touching the circumference of a circle with radius 5. Find the length of its hypotenuse.
10. If $m = \underbrace{777\dots77}_{99 \text{ digits}}$ and $n = \underbrace{999\dots99}_{77 \text{ digits}}$ multiplied together then what is the sum of digits of resulting number.

Group – B

11. If $f(n+1) = \frac{2f(n)+1}{2}$, $n=1,2,3,\dots$ and $f(1) = 2$ then find $f(101) = ?$
12. Find the remainder when $2005^{2002} + 2002^{2005}$ divided by 2003.
13. From the figure find $m\angle A + m\angle B + m\angle C + m\angle D + m\angle E + m\angle F + m\angle G = ?$

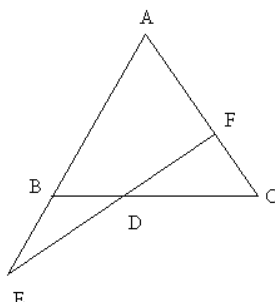


14. Triangle ABC is an acute angled triangle with $m\angle A = 30^\circ$. H is the point of intersection of altitudes from the vertices A, B and C. Midpoint of BC is M. T lies on side HM such that $HM = MT$. Prove that $AT = 2BC$.

15. Let $0 < a, b, c < 1$ and $a + b + c = 2$, prove that $\frac{a}{1-a} \cdot \frac{b}{1-b} \cdot \frac{c}{1-c} \geq 8$.

16. Find the real roots of the equation $\frac{19x+16}{10} = \left\lfloor \frac{4x+7}{3} \right\rfloor$, where $\lfloor x \rfloor$ means greatest integer value less than equal to x.

17. From the given figure $\frac{BD}{DC} = \frac{3}{4}$ and $6BE = AE$, prove that $2AF = 9CF$.



18. By using 0, 1, 2, 3, 5 without repetition of digits how many numbers lies between 100 to 1000 and how many of them are divisible by 6?

19. Find by which highest power of 2 the product of first 100 natural numbers is divisible.

20. If $x + y + z = 1$, $x^2 + y^2 + z^2 = 2$, $x^3 + y^3 + z^3 = 3$, find value of xyz .

(BONUS QUESTION)

21. Prove that there is no integral solution of the equation $1 + x^2 = 3y$.

22. There is one $1 \times n$ chess board. If each square of chess board colored with red or white or blue such that no two consecutive squares having same color. Let we can do it in a_n different ways. Find the relation between a_n , a_{n-1} and a_{n-2} .

23. Find out for what natural number n, $16n^4 + 1$ is not a prime number?

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